Project 14: PHP CI/CD Pipeline - Updated Submission

GitHub Repository URL:
https://github.com/123Origami/StegHub_DevOps/blob/main/project13

Project Overview:
This project implements a complete CI/CD pipeline for PHP applications using Jenkins, Ansible, Artifactory, and SonarQube. The pipeline automates code quality checks, testing, artifact management, and multi-environment deployments.

Key Features Implemented:

1. Infrastructure Setup:

- Jenkins server with Blue Ocean interface
- SonarQube for code quality analysis
- JFrog Artifactory for artifact management
- Multi-environment setup (Dev, SIT, Pentest)
- Nginx reverse proxy configuration

2. CI/CD Pipeline:

- Automated GitHub webhook triggers
- PHP unit test execution
- SonarQube quality gates integration
- Artifact versioning and storage
- Ansible-based deployment automation
- Multi-branch pipeline support

3. CodeQuality:

- Static code analysis with SonarQube
- Code coverage reporting
- Quality gate enforcement
- Plot visualization of metrics

4. Deployment Automation:

- Infrastructure as Code with Ansible
- Environment-specific configurations
- Database migration automation

- Zero-downtime deployment approach

Technologies Used:

- **CI/CD:** Jenkins, Blue Ocean, GitHub Webhooks
- **Code Quality:** SonarQube, PHPUnit, PHPLOC
- **Artifact Management:** JFrog Artifactory
- **Configuration Management:** Ansible
- **Infrastructure:** AWS EC2, Ubuntu 20.04
- **Web Servers:** Nginx, PHP-FPM
- **Database:** MySQL
- **Monitoring:** Nginx logs, Jenkins monitoring

Challenges Overcome:

1. **Multi-tool integration** between Jenkins, SonarQube, and Artifactory
2. **Environment-specific deployments** with Ansible inventories
3. **Quality gate implementation** to enforce code standards
4. **SSH key management** for secure Jenkins-to-server communication
5. **Database migration automation** within the pipeline